```
import torch
import torch.nn as nn


sentences = ["i love deep learning", "i love ai"]
words = list(set(" ".join(sentences).split()))
w2i = {w:i for i,w in enumerate(words)}
i2w = {i:w for w,i in w2i.items()}

data = []
for s in sentences:
    tokens = s.split()
    for i in range(len(tokens)-1):
        data.append((w2i[tokens[i]], w2i[tokens[i+1]]))

X = torch.tensor([d[0] for d in data])
Y = torch.tensor([d[1] for d in data])

class SimpleTransformer(nn.Module):
    def __init__(self, vocab_size):
        super().__init__()
        self.embed = nn.Embedding(vocab_size, 32)
        encoder_layer = nn.TransformerEncoderLayer(d_model=32, nhead=2)
        self.transformer = nn.TransformerEncoder(encoder_layer, num_layers=2)
        self.fc = nn.Linear(32, vocab_size)

    def forward(self, x):
        x = self.embed(x).unsqueeze(1)
        x = self.transformer(x)
        return self.fc(x.squeeze(1))

model = SimpleTransformer(len(w2i))

optimizer = torch.optim.Adam(model.parameters(), lr=0.01)
loss_fn = nn.CrossEntropyLoss()

for epoch in range(200):
    out = model(X)
    loss = loss_fn(out, Y)

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

test_word = "i"
test_idx = torch.tensor([w2i[test_word]])
pred = model(test_idx)
print("Next word after 'i':", i2w[torch.argmax(pred).item()])
```

```
/tmp/ipykernel_1382/353080798.py:24: UserWarning: enable_nested_tensor is True, but sel
  self.transformer = nn.TransformerEncoder(encoder_layer, num_layers=2)
Next word after 'i': love
```

```
from transformers import pipeline

fill_mask = pipeline("fill-mask", model="bert-base-uncased")
```

```
result = fill_mask("I love [MASK] learning.")
print(result[0])
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https:/
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public mode
  warnings.warn(
config.json: 100%  570/570 [00:00<00:00, 28.1kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests t
model.safetensors: 100%  440M/440M [00:03<00:00, 199MB/s]
Loading weights: 100%  202/202 [00:00<00:00, 591.29it/s, Materializing param=cls.predictions.transform.dens

BertForMaskedLM LOAD REPORT from: bert-base-uncased
Key                          | Status     |  |
-----------------------------+------------+--+-
bert.pooler.dense.weight     | UNEXPECTED |  |
bert.pooler.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight  | UNEXPECTED |  |
cls.seq_relationship.bias    | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok i
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 1.61kB/s]
vocab.txt: 232k/? [00:00<00:00, 3.16MB/s]
tokenizer.json: 466k/? [00:00<00:00, 1.52MB/s]
{'score': 0.167005717754364, 'token': 2055, 'token_str': 'about', 'sequence': 'i love a
```

```
#Next Word prediction
result = fill_mask("I love deep [MASK].")
print(result[0]['sequence'])
```

```
i love deep places.
```

```
#Review Classification
classifier = pipeline("sentiment-analysis")

print(classifier("This product is amazing!"))
print(classifier("Worst experience ever"))
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-
Using a pipeline without specifying a model name and revision in production is not recc
config.json: 100%  629/629 [00:00<00:00, 25.4kB/s]
model.safetensors: 100%  268M/268M [00:01<00:00, 274MB/s]
Loading weights: 100%  104/104 [00:00<00:00, 394.91it/s, Materializing param=pre_classifier.weight]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 2.43kB/s]
vocab.txt: 232k/? [00:00<00:00, 7.07MB/s]
[{'label': 'POSITIVE', 'score': 0.9998860359191895}]
[{'label': 'NEGATIVE', 'score': 0.9997971653938293}]
```

```
pip install torchvision transformers
```

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pip install torchvision matplotlib
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```

```
from transformers import ViTImageProcessor, ViTForImageClassification
from PIL import Image
import requests
model = ViTForImageClassification.from_pretrained("google/vit-base-patch16-224")
processor = ViTImageProcessor.from_pretrained("google/vit-base-patch16-224")
url = "https://images.unsplash.com/photo-1518791841217-8f162f1e1131"
image = Image.open(requests.get(url, stream=True).raw)
inputs = processor(images=image, return_tensors="pt")
outputs = model(**inputs)
predicted_class = outputs.logits.argmax(-1).item()

print("Predicted class:", model.config.id2label[predicted_class])
```

```
config.json:        69.7k/?  [00:00<00:00,  1.39MB/s]

model.safetensors:  100%                                          346M/346M  [00:04<00:00,  131MB/s]

Loading weights:  100%          200/200  [00:00<00:00,  330.10it/s,  Materializing  param=vit.layernorm.weight]

preprocessor_config.json:  100%                                   160/160  [00:00<00:00,  1.80kB/s]
Predicted class: tabby, tabby cat
```

```
import torch
import torch.nn as nn
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
transform = transforms.ToTensor()
dataset = torchvision.datasets.MNIST(root='./data', download=True, transform=transform)
loader = torch.utils.data.DataLoader(dataset, batch_size=64, shuffle=True)


class Generator(nn.Module):
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Linear(100, 128),
            nn.ReLU(),
            nn.Linear(128, 784),
            nn.Tanh()
        )
```

```
    def forward(self, x):
        return self.model(x)


class Discriminator(nn.Module):
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Linear(784, 128),
            nn.ReLU(),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )

    def forward(self, x):
        return self.model(x)

G = Generator()
D = Discriminator()

loss_fn = nn.BCELoss()
opt_G = torch.optim.Adam(G.parameters(), lr=0.0002)
opt_D = torch.optim.Adam(D.parameters(), lr=0.0002)


for epoch in range(3):
    for real, _ in loader:
        real = real.view(-1, 784)


        z = torch.randn(real.size(0), 100)
        fake = G(z)

        loss_D = loss_fn(D(real), torch.ones(real.size(0),1)) + \
                 loss_fn(D(fake.detach()), torch.zeros(real.size(0),1))

        opt_D.zero_grad()
        loss_D.backward()
        opt_D.step()

        z = torch.randn(real.size(0), 100)
        fake = G(z)

        loss_G = loss_fn(D(fake), torch.ones(real.size(0),1))

        opt_G.zero_grad()
        loss_G.backward()
        opt_G.step()

    print(f"Epoch {epoch} Done")


z = torch.randn(1, 100)
generated = G(z).view(28,28).detach()

plt.imshow(generated, cmap='gray')
plt.title("Generated Digit")
plt.show()
```